

1 Metric TSP and Eulerian Structure

1.1 General TSP, metric TSP, and the shortcut rule

The traveling salesperson problem asks for a minimum-cost Hamiltonian cycle: visit every vertex exactly once and return to the start. Throughout the approximation algorithms below, the input is a complete undirected graph with nonnegative symmetric costs satisfying the triangle inequality

$$c(u, w) \leq c(u, v) + c(v, w).$$

These are *metric* costs. Completeness supplies each shortcut edge; triangle inequality ensures that shortcutting a walk never increases its cost. Repeatedly bypass already visited vertices, retaining the final return to the start, to turn a spanning closed walk into a Hamiltonian cycle.

Why the metric assumption matters. General TSP has no polynomial-time approximation with any fixed finite factor $\rho \geq 1$, unless $P = NP$. Reduce Hamiltonian Cycle on an unweighted graph G with n vertices: complete the graph, give original edges cost 1 and nonedges cost an integer $B > \rho n$. A YES instance has $\text{OPT} = n$, so a ρ -approximation returns cost at most ρn . In a NO instance every tour uses a nonedge, hence has cost at least $B > \rho n$. The algorithm would distinguish the two cases in polynomial time. For fixed ρ , B has polynomial encoding length. These constructed costs need not satisfy triangle inequality; a cost- B shortcut can replace a two-edge detour of cost 2.

Metric TSP remains NP-hard: assign costs 1 to edges of a Hamiltonian Cycle instance and 2 to nonedges. Every triangle satisfies the metric inequality, and a tour of cost n exists exactly when the original graph is Hamiltonian. This proves exact hardness, not the absence of constant-factor approximation.

1.2 Euler circuits, open trails, and orientation

An Euler trail uses every *edge occurrence* exactly once; vertices may repeat. Parallel edges are distinct occurrences. An Euler circuit is a closed Euler trail. A Hamiltonian cycle instead visits every vertex once.

Theorem 1.1 (Euler's criterion). An undirected multigraph with at least one edge has an Euler circuit iff all its nonisolated vertices are connected and every vertex has even degree. It has an Euler trail between distinct vertices s, t iff all nonisolated vertices are connected and precisely s, t have odd degree.

Proof. In a closed traversal, each arrival at a vertex is paired with a departure, so every degree is even. For an open traversal only the start and end have one unpaired incident edge.

Conversely, start at any nonisolated vertex and keep following unused edges. An even-degree vertex other than the start cannot be the first place where this trail gets stuck: after arriving, an unused departure remains. Thus the trail closes. If edges remain, connectivity gives a vertex of the current circuit incident to unused edges. The unused-edge degrees are still even, so construct another closed trail there and splice it into the first. Repeat until every edge is used (Hierholzer's algorithm). For the open case add one extra $\{s, t\}$ edge, find an Euler circuit, rotate it to place that edge last, and delete the extra edge. $\square$

Using adjacency lists and marking each edge occurrence, Hierholzer's algorithm runs in $O(|V| + |E|)$ time. The handshaking identity $\sum_v \deg(v) = 2|E|$ also shows that the number of odd-degree vertices is even. Adding one edge toggles the parity of its two endpoints.

Orienting an even-degree connected graph. Find an Euler circuit and orient each edge in the direction in which the circuit traverses it. The resulting directed graph is strongly connected: from any vertex u , follow the directed circuit cyclically until any desired vertex v is reached. Also $\deg^+(v) = \deg^-(v)$ at every vertex. This proof concerns the nonisolated vertices; a disconnected even-degree graph does not become strongly connected by orienting its components.

1.3 Double-tree: a 2-approximation

1. Compute a minimum spanning tree T .
2. Duplicate every edge of T , retaining both copies in a multigraph.
3. Find an Euler circuit in this connected even-degree multigraph.
4. Shortcut repeated vertex visits to obtain the output tour H .

Feasibility. Doubling makes every degree even; the spanning tree ensures connectivity. The Euler circuit visits all vertices, and shortcutting yields a Hamiltonian cycle in the complete graph.

Lower bound and guarantee. Deleting any edge of an optimal tour leaves a spanning tree, so $c(T) \leq \text{OPT}$. Therefore

$$c(H) \leq 2c(T) \leq 2\text{OPT}.$$

The first inequality uses triangle inequality; the second uses the MST lower bound. Dense Prim takes $O(n^2)$ time on the complete graph; doubling, Euler traversal, and shortcutting take $O(n)$ further time.

Four-vertex metric counterexample to optimality. Use the shortest-path metric of the unit-cost cycle A, B, D, C, A :

c	A	B	C	D
A	0	1	1	2
B	1	0	2	1
C	1	2	0	1
D	2	1	1	0

The tree $\{AB, AC, CD\}$ is an MST of cost 3. The doubled tree admits Euler circuit A, B, A, C, D, C, A . First-visit shortcutting gives A, B, C, D, A , of cost $1 + 2 + 1 + 2 = 6$. The original cycle has cost 4 and is optimal because every tour has four edges, each of cost at least 1. Thus a valid double-tree execution need not be optimal, even on a metric.

1.4 Christofides: a 3/2-approximation

1. Compute an MST T , and let $O = \{v : \deg_T(v) \text{ is odd}\}$.
2. Find a *minimum-weight perfect matching* M on the complete graph induced by O , using the original metric costs.
3. Form the multigraph $T \cup M$, find an Euler circuit, and shortcut repeated vertices to produce a tour H .

Feasibility. The set O has even cardinality. Every vertex in O gains exactly one incident matching edge and becomes even; all other degrees stay even. The tree ensures connectivity. Keep parallel copies if a matching edge is also in T .

Lemma 1.2 (Matching budget). The matching chosen above has cost $c(M) \leq \text{OPT}/2$.

Proof. Take an optimal tour and shortcut it to retain only the vertices of O . The resulting even-length cyclic walk has cost at most OPT . Its alternating edges split into two perfect matchings M_1, M_2 on O . Therefore, by minimum weight of M ,

$$c(M) \leq \min\{c(M_1), c(M_2)\} \leq \frac{c(M_1) + c(M_2)}{2} \leq \frac{\text{OPT}}{2}.$$

If O has two vertices, the restricted cyclic walk traverses their edge twice; its two edge occurrences give the same one-edge matching. If $O = \emptyset$, use the empty matching. $\square$

Combining the spanning and matching budgets with metric shortcutting gives

$$c(H) \leq c(T) + c(M) \leq \text{OPT} + \frac{1}{2} \text{OPT} = \frac{3}{2} \text{OPT}.$$

All steps are polynomial time; minimum-weight perfect matching is the main additional subroutine beyond double-tree. A maximal matching is insufficient: it may leave odd vertices unmatched, and an arbitrary perfect matching has no $\text{OPT}/2$ cost guarantee.

1.5 Held–Karp exact dynamic programming

This exact algorithm works for arbitrary tour costs; triangle inequality is unnecessary. Fix a root r . For $r \in S \subseteq V$ and $v \in S$, let $D[S, v]$ be the minimum cost of a path starting at r , ending at v , and visiting exactly the vertices in S , once each.

Bases. $D[\{r\}, r] = 0$. Set all impossible states to ∞ , including $D[S, r]$ for $|S| > 1$ and states with $r \notin S$ or $v \notin S$. For $v \neq r$,

$$D[S, v] = \min_{u \in S \setminus \{v\}} (D[S \setminus \{v\}, u] + c(u, v)).$$

Why the recurrence is exact. The penultimate vertex u partitions all feasible paths. Removing their last edge leaves precisely a feasible path for the smaller state; conversely, appending (u, v) to any such path gives a feasible path for (S, v) .

Evaluate states in increasing $|S|$. For $n \geq 2$, close the path by returning to r :

$$\text{OPT} = \min_{v \neq r} (D[V, v] + c(v, r)).$$

Store a minimizing predecessor with each state; choose the minimizing final vertex, backtrack while removing the last vertex from S , reverse the resulting sequence, and append r . There are $O(n2^n)$ states with $O(n)$ transitions each, giving $O(n^2 2^n)$ time and $O(n2^n)$ space, including predecessors. This is exponential, although faster than enumerating tours.

2 Targeted Practice Questions

These practice variations combine the supplied lectures, Tutorials 1–4, and question styles from the two sample finals. They are speculative revision questions, not information about the actual midterm. Marks are suggested practice weights. Attempt the questions before consulting the separate answer sketches.

2.1 Questions

Q1. Scheduling with two levels of quotas (10 marks). There are workers i , disjoint periods j , and tasks, each belonging to one period. Worker i can perform tasks in A_i , may perform at most b_i tasks overall, and at most one task per period. Every task must be performed by exactly one worker. (a) Construct a flow network and specify the target flow value. (b) Prove both directions of the reduction. (c) Give an ILP. (d) Explain why a fractional feasible flow is not itself a schedule, and why this does not prevent a flow-based algorithm.

Q2. Bounded-frequency weighted covering (8 marks). Each element belongs to at most k candidate sets with nonnegative costs. Give an ILP and a deterministic k -approximation using its LP relaxation. Prove feasibility and cost separately. Does the same threshold proof work if instead every set contains at most k elements?

Q3. Randomized cut, then remove the randomness (8 marks). For weighted MAX- k -CUT on a loopless undirected graph with nonnegative weights, design an algorithm obtaining at least $(1 - 1/k)$ OPT in expectation. Convert it to a deterministic polynomial-time algorithm. State precisely how to compute conditional expectation after a partial labelling. Explain why independence of the edge indicators is unnecessary.

Q4. Randomized set cover with a stopping rule (10 marks). Solve the natural weighted-set-cover LP on a feasible instance with $n \geq 1$ elements. One trial takes $r = \lceil \ln(2n) \rceil$ independent rounds, selecting each set j independently with probability x_j^* in each round and taking their union. Discard unsuccessful trials and repeat independently until a cover is obtained. Prove almost-sure feasibility, expected polynomial running time, and expected returned cost at most $2r$ OPT. Explain why “one trial has expected cost at most rLP^* , so the returned cover does too” is not justified.

Q5. Minimum-cut structure and tie-breaking (8 marks). (a) Fix any maximum flow f . Prove that for *every* minimum cut, outgoing arcs are saturated and incoming arcs carry zero flow. (b) Deduce that a positive-capacity arc cannot cross two minimum cuts in opposite directions. (c) For integer capacities, find a minimum cut with the fewest outgoing arcs.

Q6. MAX-DICUT rounding and a changed probability (8 marks). For a loopless directed graph with nonnegative arc weights, use the LP $z_{ij} \leq x_i$, $z_{ij} \leq 1 - x_j$, with variables in $[0, 1]$, maximizing $\sum_{ij} w_{ij} z_{ij}$. Prove that independently placing vertex i in the source side with probability $p_i = 1/4 + x_i/2$ gives at least half the LP value in expectation. As an extension, determine whether direct rounding $p_i = x_i$ also guarantees half the optimum when the LP solution is optimal.

Q7. MAX-SAT clause lengths and combination (10 marks). Write the weighted MAX-SAT ILP and LP. Prove the LP-rounding bound for a clause of length ℓ , including the AM–GM and concavity steps. Then show why taking the better of LP rounding and fair random assignment obtains $3 \text{ OPT} / 4$ in expectation. State the preprocessing needed for the clause-product calculation.

Q8. Metric TSP and the role of assumptions (8 marks). Prove the double-tree factor 2 and Christofides factor $3/2$. Explain exactly where triangle inequality, even parity, and minimum-weight perfect matching are used. Give a four-vertex metric instance where a valid double-tree execution is nonoptimal. Explain why arbitrary nonmetric costs invalidate the shortcut argument.

Q9. Knapsack scaling (8 marks). For nonnegative integer item values, derive the value-indexed $O(n^2V)$ DP, where V is the largest item value. Use it to obtain an FPTAS with $O(n^3/\epsilon)$ arithmetic-operation complexity. Prove the guarantee and explain why overweight items must be removed before defining V .

Q10. Vertex-disjoint routes (8 marks). For distinct nonadjacent vertices x, y of an undirected graph, prove that the maximum number of internally vertex-disjoint x - y paths equals the minimum size of an x - y vertex separator. Give the flow construction and justify both correspondences. Why do capacity-1 splitting arcs suffice, and why are the other arcs assigned capacity larger than n ?

Q11. Repair a flow algorithm (8 marks). (a) A scaling algorithm starts at $k = C$, halves by real division, and stops below 1. Find an integer-capacity counterexample and repair the algorithm. (b) Prove that BFS augmentations give $O(nm^2)$ time independent of capacity magnitudes. (c) Explain why arbitrary Ford–Fulkerson is only pseudo-polynomial for integer capacities and may fail to terminate for irrational capacities.

Q12. An unfamiliar objective with a familiar proof (6 marks). You are given n strings of length L over an alphabet of q symbols. Give a deterministic $1/q$ retained-value approximation for their longest common subsequence in $O(nL + q)$ time. Give a second proof using a sum of per-symbol upper bounds on OPT.

2.2 Quick true/false drill

For each statement give a proof or a small counterexample.

- T1.** Complementing any 2-approximate vertex cover gives a constant-factor approximate maximum independent set.
- T2.** Integer capacities imply every maximum flow is integral on every arc.
- T3.** A feasible flow whose value equals the capacity of some cut is maximum.
- T4.** An integer-capacity flow algorithm taking $O(mC)$ time is polynomial in the binary input length.
- T5.** A constant-factor approximation guarantee for MAX-SAT automatically gives one for minimizing unsatisfied clauses.
- T6.** Shortest residual distances never decrease under arbitrary Ford–Fulkerson path choices.
- T7.** A feasible solution of a minimization LP relaxation upper-bounds the integer optimum.
- T8.** Every graph on $n \geq 3$ vertices has vertex-cover LP integrality gap at least $2 - 2/n$.
- T9.** Endpoints of a maximal matching always form a vertex cover; endpoints of a maximum matching always form a minimum cover.
- T10.** The indicators of two edges sharing a vertex must be independent in order to sum their expectations.

2.3 Answer sketches

Q1. Scheduling. Use layers

$$s \longrightarrow i \longrightarrow (i, j) \longrightarrow d \longrightarrow t$$

with capacities $b_i, 1, 1, 1$, respectively. Include $(i, j) \rightarrow d$ exactly when task d is in period j and $d \in A_i$. The target flow value is the number D of tasks. A schedule sends one unit on the path for each assignment. Conversely, an integral flow of value D saturates all task-to-sink arcs. Each task then has exactly one unit of incoming flow; decode that arc as its assigned worker. The worker and worker-period capacities enforce the two quotas. Fractional flow can split a task across workers, but integer capacities guarantee the existence of an integral maximum flow, which the standard augmenting-path algorithms can return.

For available pairs use binary a_{id} and the feasibility ILP

$$\sum_{i: d \in A_i} a_{id} = 1 \quad (\text{each } d), \quad \sum_{d \in A_i} a_{id} \leq b_i \quad (\text{each } i), \quad \sum_{\substack{d \in A_i \\ d \text{ in period } j}} a_{id} \leq 1 \quad (\text{each } i, j).$$

The network has polynomial size and requires one maximum-flow computation.

Q2. Bounded-frequency covering. Minimize $\sum_j c_j x_j$ subject to $\sum_{j: e \in S_j} x_j \geq 1$ for every element, and $x_j \in \{0, 1\}$. Relax to $[0, 1]$ and select $x_j^* \geq 1/k$. A covering constraint has at most k terms, so at least one meets the threshold. With nonnegative costs,

$$\text{ALG} \leq \sum_j k c_j x_j^* = k \text{LP}^* \leq k \text{OPT}.$$

Bounded *set size* does not bound the number of terms in a covering constraint. For example, one element and two identical unit-cost singleton sets admit optimal LP solution $(1/2, 1/2)$. Set size is at most 1, but threshold 1 selects neither set.

Q3. MAX- k -CUT and conditional expectation. Independent uniform vertex labels cut an edge with probability $1 - 1/k$. Linearity gives expectation $(1 - 1/k) \sum_e w_e \geq (1 - 1/k) \text{OPT}$; edge indicators need not be independent. Under a partial labelling, an edge with both endpoints fixed contributes its weight or zero. An edge with at least one endpoint unfixed contributes $(1 - 1/k)w_e$. For the next vertex, only edges to already fixed neighbors have a label-dependent contribution. Choose the label maximizing the weight of differently labelled fixed neighbors. The current conditional expectation is the average of the k branch expectations, so this choice cannot decrease it. After all choices the objective equals its conditional expectation. A direct implementation takes $O(k(n + m))$ time.

Q4. Set cover with rejection sampling. For any element e , one round misses it with probability

$$\prod_{j: e \in S_j} (1 - x_j^*) \leq \exp\left(-\sum_{j: e \in S_j} x_j^*\right) \leq e^{-1}.$$

After r rounds this is at most $e^{-r} \leq 1/(2n)$. A union bound implies trial success probability $p \geq 1/2$. For trial cost C ,

$$\mathbb{E}[C] = \sum_j c_j (1 - (1 - x_j^*)^r) \leq r \sum_j c_j x_j^* = r \text{LP}^*.$$

The accepted trial has the distribution of one trial conditioned on success A . Since $C \geq 0$,

$$\mathbb{E}[C \mid A] = \frac{\mathbb{E}[C \mathbf{1}_A]}{p} \leq \frac{\mathbb{E}[C]}{p} \leq 2r \text{LP}^* \leq 2r \text{OPT}.$$

Fresh independent trials give a geometric stopping time of mean $1/p \leq 2$; nontermination probability is $\lim_{t \rightarrow \infty} (1 - p)^t = 0$. Each trial is polynomial time and its output is checked for feasibility. The success event may be correlated with cost, so the unconditioned trial expectation alone does not bound the returned cost.

Q5. Cuts. For a fixed maximum flow and any minimum-cut source side S ,

$$0 = c(\delta^+(S)) - |f| = \sum_{e \in \delta^+(S)} (c_e - f_e) + \sum_{e \in \delta^-(S)} f_e.$$

All summands are nonnegative, so each vanishes. An arc with $c_e > 0$ crossing outwards forces $f_e = c_e > 0$; the same arc crossing inwards for another minimum cut would force $f_e = 0$, a contradiction. For tie-breaking let $m = |E|$ and set $c'_e = (m+1)c_e + 1$. Then

$$c'(\delta^+(S)) = (m+1)c(\delta^+(S)) + |\delta^+(S)|.$$

An integer capacity difference of at least 1 outweighs every cardinality difference, which is at most m . Minimize the new capacity with one max-flow/min-cut computation. Its encoding length grows by $O(\log(m+1))$.

Q6. MAX-DICUT. By independence for the two distinct endpoints and the LP constraints,

$$\begin{aligned} \Pr[i \in U, j \notin U] &= (1/4 + x_i/2)(1/4 + (1 - x_j)/2) \\ &\geq (1/4 + z_{ij}/2)^2 = z_{ij}/2 + (2z_{ij} - 1)^2/16 \geq z_{ij}/2. \end{aligned}$$

Sum with the nonnegative weights. For direct rounding, put $L = \sum w_{ij}z_{ij} = LP^*$ and $W = \sum w_{ij}$. If $W = 0$ the claim is trivial. Otherwise $x_i(1 - x_j) \geq z_{ij}^2$, and Cauchy-Schwarz gives

$$\mathbb{E}[\text{ALG}] \geq \sum w_{ij}z_{ij}^2 \geq L^2/W.$$

The feasible point $x_i = z_{ij} = 1/2$ implies $L \geq W/2$ when the LP solution is optimal. Hence $\mathbb{E}[\text{ALG}] \geq L/2 \geq \text{OPT}/2$. The shifted rule has a local per-arc proof; the direct rule uses LP optimality for its global comparison.

Q7. MAX-SAT. Remove duplicate literals; treat tautological clauses as always satisfied. For each remaining nonempty clause C_i , use

$$\max \sum_i w_i y_i, \quad \sum_{j \in C_i^+} x_j + \sum_{j \in C_i^-} (1 - x_j) \geq y_i, \quad x_j, y_i \in \{0, 1\};$$

relax the domains to $[0, 1]$. Independent variable rounding gives literal success probabilities $a_1, \dots, a_\ell$ in a clause with distinct variables, and $\sum_h a_h \geq y_i^*$. AM-GM gives

$$\Pr[C_i \text{ fails}] = \prod_h (1 - a_h) \leq \left(1 - \frac{\sum_h a_h}{\ell}\right)^\ell \leq (1 - y_i^*/\ell)^\ell.$$

The function $g(y) = 1 - (1 - y/\ell)^\ell$ is concave on $[0, 1]$ with $g(0) = 0$. Thus $g(y) \geq yg(1)$, and the clause contributes at least $b_\ell w_i y_i^*$ where $b_\ell = 1 - (1 - 1/\ell)^\ell \geq 1 - 1/e$. Fair assignment contributes at least $a_\ell w_i y_i^*$, with $a_\ell = 1 - 2^{-\ell}$ because $y_i^* \leq 1$.

For $\ell = 1, 2$, $a_\ell + b_\ell = 3/2$; for $\ell \geq 3$, $a_\ell \geq 7/8$ and $b_\ell \geq 1 - 1/e > 5/8$. Therefore, for outputs A, B ,

$$\mathbb{E}[\max(A, B)] \geq \frac{1}{2}(\mathbb{E}[A] + \mathbb{E}[B]) \geq \frac{3}{4}LP^* \geq \frac{3}{4}\text{OPT}.$$

The two outputs need not be independent. Averaging just the global $1/2$ and $1 - 1/e$ guarantees would not prove $3/4$.

Q8. TSP. Deleting a tour edge proves $c(T) \leq \text{OPT}$ for an MST T . Doubling gives even degrees and cost $2c(T)$; triangle inequality makes the final shortcut tour no more expensive. For Christofides, the odd-degree set has even cardinality. Shortcut an optimal tour to this set and split its alternating edges into two perfect matchings. The minimum matching M costs no more than their average, at most $\text{OPT}/2$. Thus the output costs at most $c(T) + c(M) \leq 3\text{OPT}/2$.

For the requested example, use the unit-cost cycle A, B, D, C, A and diagonal costs 2. MST $\{AB, AC, CD\}$ has Euler circuit A, B, A, C, D, C, A after doubling. First-visit shortcutting returns A, B, C, D, A , cost 6, while the original cycle has optimal cost 4. Without triangle inequality a shortcut edge can cost arbitrarily more than the detour it replaces.

Q9. Knapsack. Discard $w_i > W$, and handle no remaining items or $V = 0$ separately. Let $D[i, p]$ be the minimum weight of a subset of the first i items with value exactly p . Set $D[0, 0] = 0$, all other initial states to ∞ , and

$$D[i, p] = \min\{D[i-1, p], w_i + D[i-1, p - v_i]\},$$

omitting the second term when $p < v_i$. Fill increasing i ; choose the largest p with $D[n, p] \leq W$ and backtrack stored choices. Since $p \leq nV$, this takes $O(n^2V)$ arithmetic operations.

For $0 < \epsilon < 1$, put $K = \epsilon V/n$ and $\hat{v}_i = \lfloor v_i/K \rfloor \leq n/\epsilon$. Solve the DP with scaled values and original weights. It takes $O(n^3/\epsilon)$ operations. For its returned set S and an original optimum S^* ,

$$v(S) \geq K\hat{v}(S) \geq K\hat{v}(S^*) \geq \text{OPT} - nK = \text{OPT} - \epsilon V \geq (1 - \epsilon) \text{OPT}.$$

Filtering ensures that the maximum-value item is itself feasible, so $V \leq \text{OPT}$. Keeping original weights preserves feasibility.

Q10. Vertex form of Menger's theorem. Split each internal vertex into $v_{\text{in}} \rightarrow v_{\text{out}}$ of capacity 1, leaving x, y unsplit. Replace each undirected edge $\{u, v\}$ by $u_{\text{out}} \rightarrow v_{\text{in}}$ and $v_{\text{out}} \rightarrow u_{\text{in}}$ of capacity $M = n + 1$, interpreting an endpoint x or y as the unsplit vertex. Internally disjoint paths induce a feasible integral flow. Conversely, decompose integral flow into unit x - y paths and cycles and discard cycles; capacity-1 splitting arcs ensure internal vertex-disjointness.

A separator of size k deletes k splitting arcs and destroys all flow paths. Reachability after this deletion gives an original-network cut of capacity at most k . Because x, y are nonadjacent, all $n - 2$ internal vertices form a separator, so a minimum cut has capacity at most $n - 2 < M$. Consequently it cuts no capacity- M arc. Its outgoing splitting arcs identify a vertex separator whose size equals the cut capacity. Max-flow/min-cut now equates the two optima. Adjacency is excluded because a direct x - y edge survives deletion of every internal vertex.

Q11. Flow algorithms. A two-arc path with capacities 3, 1 has maximum flow 1, but thresholds 3, 1.5 find no path before the flawed algorithm stops. Start instead at $2^{\lceil \log_2 C \rceil}$ and halve through threshold 1; handle $C = 0$ separately. Use only residual arcs of capacity at least the current threshold.

For BFS augmentation, let $d(v)$ be residual distance from s before an augmentation. Every surviving old arc satisfies $d(v) \leq d(u) + 1$. A new arc (v, u) reverses an arc of the chosen shortest path, so $d(v) = d(u) + 1$; it too satisfies the inequality needed along any new path. Induction along a new residual path proves that no distance decreases. If (u, v) is saturated on a shortest path, then $d(v) = d(u) + 1$. Before it can be saturated in that direction again, an augmentation must use (v, u) ; at that later time

$$d_{\text{later}}(u) = d_{\text{later}}(v) + 1 \geq d_{\text{earlier}}(v) + 1 = d_{\text{earlier}}(u) + 2.$$

Thus each labelled residual arc can be critical only $O(n)$ times. Each augmentation saturates one such arc, giving $O(nm)$ augmentations and $O(nm^2)$ time under the usual $O(m)$ search bound.

With integer capacities, arbitrary Ford-Fulkerson increases value by at least 1, giving $O(m|f^*|)$ time. The numeric value $|f^*|$ may be exponential in its binary encoding length. With irrational capacities, there need be no positive lower bound on augmentation amounts; bad path choices can produce infinitely many positive augmentations. BFS avoids this dependence by bounding the number of augmentations combinatorially.

Q12. Common subsequence. For each symbol a , compute $f_a = \min_i \text{count}_a(s_i)$. Return a repeated f_a times for a symbol maximizing f_a ; this is always a common subsequence. Some symbol appears at least OPT/q times in an optimal common subsequence, and its f_a is at least that large. Alternatively,

$$\text{OPT} \leq \sum_a f_a \leq q \max_a f_a.$$

To achieve $O(nL + q)$ even for variable q , initialize arrays once; while scanning each string, use timestamps to count only its occurring symbols. For each touched symbol update its minimum count and the number of strings in which it appears. Symbols appearing in fewer than n strings get $f_a = 0$. Each character and each touched symbol costs $O(1)$, followed by an $O(q)$ final scan; no $O(q)$ reset per string is needed.

2.4 True/false answers

- T1. False.** Take q disjoint edges and one isolated vertex. Selecting all edge endpoints is a 2-approximate cover; its complement has size 1, while a maximum independent set has size $q + 1$.
- T2. False.** Use unit arcs $s \rightarrow a, s \rightarrow b, a \rightarrow c, b \rightarrow c, c \rightarrow t$. Sending $1/2$ on each branch and 1 on $c \rightarrow t$ is a fractional maximum flow. An integral maximum also exists.
- T3. True.** Weak duality gives $|f| \leq |f^*| \leq c(S)$; equality of the outer values certifies maximum flow and minimum cut.
- T4. False.** Capacity C needs only $O(\log(C + 1))$ bits. A bound polynomial in C may be exponential in input length.
- T5. False.** A satisfiable formula can have minimum unsatisfied cost 0. An assignment satisfying only half of its clauses may meet a MAX-SAT factor-2 guarantee while leaving positive cost, impossible for any finite multiplicative guarantee relative to 0.
- T6. False.** Use unit arcs along $s \rightarrow a \rightarrow b \rightarrow c \rightarrow d \rightarrow t$ and an additional arc $s \rightarrow d$. Initially $d(s, c) = 3$. Augment on the long path: the unused arc $s \rightarrow d$ and new reverse arc $d \rightarrow c$ give distance 2 to c .
- T7. False.** The optimal relaxation value is a *lower* bound. An arbitrary fractional feasible value need not bound the integral optimum in either direction. An integral feasible solution does give an upper bound.
- T8. False.** K_n witnesses a worst-case gap $2 - 2/n$: integral optimum $n - 1$, LP optimum $n/2$. This is not a statement about every graph; a single edge with isolated vertices has gap 1.
- T9. First true, second false.** An uncovered edge could be added to a matching, contradicting maximality. For one edge the maximum matching's endpoints give cover size 2, but minimum cover size 1.
- T10. False.** Linearity of expectation requires no independence.